Does the NIRF research score track what researchers publish?

Every year NIRF ranks India’s top 100 institutions, and part of the score rewards research. We checked it against an independent source: OpenAlex papers and citations, 2017 to 2025.

Bhanu Manideep
DA25B024

Sasanka Mouli
DA25B013

Kushal V
DA25B019

Two sources, joined through ROR

Source	What we took	Licence or terms	Collected
NIRF Overall ranking pages, nirfindia.org/Rankings/YEAR/OverallRanking.html	Rank and five scores, 2017 to 2025	Public government pages, no terms page	2026-10-31
OpenAlex API, api.openalex.org	Papers and citations per institution and year	CC0, public domain	2026-11-01
ROR API v2, api.ror.org	Registry ids that link NIRF names to OpenAlex	CC0 1.0	2026-11-01

How we compare

NIRF year Y against OpenAlex output over the three years before it, Y–3 to Y–1, counting only the institution’s own papers.

The measure

Spearman rank correlation per year with a 95% bootstrap interval: 1 means the same order, 0 means unrelated.

What we ended up with

164 institutions, 9 ranking years, 901 institution-years, each with 3 years of OpenAlex counts.

Collected politely, and checked first

robots.txt

We asked for it before scraping. NIRF has none: the server answered with an error page, “IIS 10.0 Detailed Error – 404.0 – Not Found”, on 2026-10-31. No crawl rules are published.

Terms of use

The NIRF site has no separate terms page. Under the NIRF methodology, institutions must publish the data they submit. We treat the tables as public records, use them for academic work and name NIRF as the source.

Politeness

One request at a time, 1.5 seconds between requests, nine pages in total. OpenAlex and ROR were called through their public APIs, with a pause between calls and well below their limits.

Raw files are never edited

The 9 NIRF pages (2017 to 2025) and 1,932 cached OpenAlex answers sit in raw/ as downloaded. Every later step reads them and writes somewhere else.

Raw, interim, clean: each step leaves a file

raw/

Pages and JSON exactly as downloaded. Never edited.

interim/

Parsed tables, the ROR crosswalk, tests and the cleaning logs.

clean/

The joined table the dashboard reads, plus the data dictionary.

data/

One SQLite file of the joined table.

File names carry source and date, for example nirf_openalex_joined_2026-11-01.csv.

CSV for tables

Plain text, opens anywhere, easy to compare between runs. Every notebook and the dashboard read it.

SQLite for the joined table

One file, no server, and the dashboard asks it questions with SQL. Read back, it has the same shape, values, empty cells and dtypes as the CSV.

A data dictionary

For every column: dtype, unit, source, allowed range and the range actually observed.

Cleaning: fewer names, no rows lost

342 → 164
institution ids

231 → 164
name spellings

106 → 92
city spellings

901 → 901
rows, none dropped

Missing values

No empty scores. But 90 of 801 institutions vanish from next year's list, and they ranked lower (mean rank 81.1 against 46.7, permutation p = 0.0001). That is MAR, not random, so we keep observed rows and impute nothing.

Outliers

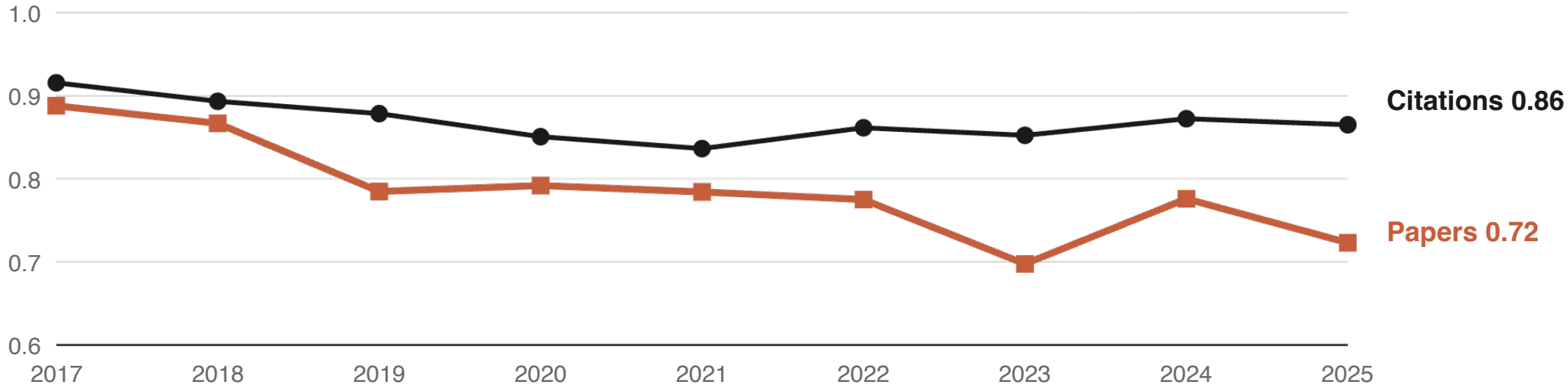
IQR on log papers, modified z and Mahalanobis flag 7 institution-years. Four have very few papers and three have very many. All are real institutions, so we keep them and rerun the result without them.

Duplicates

0 exact repeats. 8 pairs of near-identical names, such as IIT campuses, are different places. 15 NIRF–OpenAlex name mismatches are renamed institutions, checked against ROR city and type.

It tracks: citations closely, paper counts less so

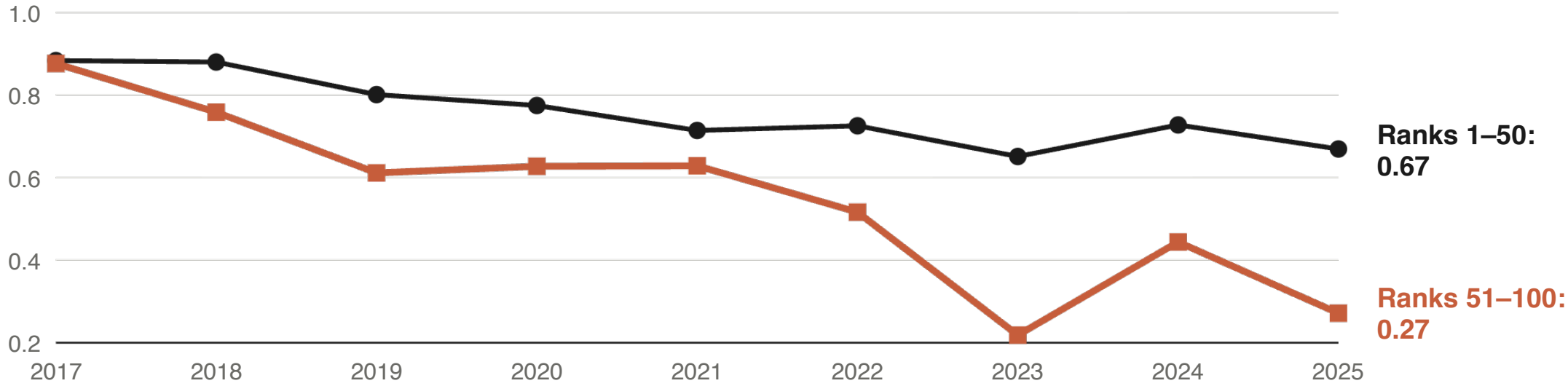
Spearman rank correlation with the NIRF research score (1 = same order)



In 2025 the research score and OpenAlex citations agree at 0.86 (95% interval 0.80 to 0.91); with paper counts the figure is 0.72 (0.61 to 0.80). Citations stay between 0.84 and 0.92 in all nine years. Paper counts began at 0.89 in 2017 and have drifted down.

The link has faded in the bottom half

Rank correlation with OpenAlex paper counts (1 = same order), by NIRF rank band



Among institutions ranked 51 to 100, agreement with paper counts fell from 0.88 in 2017 to 0.27 in 2025, while ranks 1 to 50 stayed at 0.65 or higher. The share of highly cited papers (10 or more citations) matches the score at only 0.44 in 2025. In 2017 NIRF's research score put JNCASR 11th, yet it was 60th of 100 on papers.

So, does it track?

Mostly yes. The ranking order is close to the output order.

0.86

with OpenAlex citations in 2025, and between 0.84 and 0.92 in every year.

0.72

with paper counts in 2025, down from 0.89 in 2017.

0.27

for ranks 51 to 100 in 2025, down from 0.88 in 2017.

The research score is a dependable guide to who sits near the top. It says much less about the bottom half of the list, and the link has loosened since 2017.

What this cannot tell you

Agreement is not cause

Rank correlation shows the orders match. The NIRF research score also counts patents and funded projects, so it was never meant to equal publications.

Only institutions on the list

Those who drop out ranked lower (mean rank 81.1 against 46.7). Results hold for the top 100, not for all Indian institutions.

Matching names to records

OpenAlex counts can include affiliated units. In 88 institution-years the broader count ran more than 5% above the institution's own, so we used the own count.

Choices we made

A three-year window. Shifting it by one year moves any correlation by at most 0.06, so the story holds, but the exact values depend on it.

A snapshot

Collected on 2026-10-31 and 2026-11-01. OpenAlex keeps adding records, and 7 perception scores of exactly 0 are kept as published.

Next step

Compare citations adjusted for field, and test each NIRF score component against its own OpenAlex counterpart.

Credits

Member	Roll number	Contribution
Bhanu Manideep	DA25B024	ROR and OpenAlex collection, data cleaning
Sasanka Mouli	DA25B013	NIRF scraping, data cleaning
Kushal V	DA25B019	Visualisation and dashboard

Data and licences

NIRF Overall rankings, nirfindia.org, public pages. OpenAlex, CC0. ROR, CC0 1.0.

Fonts

Libre Franklin and Source Serif 4, SIL Open Font License.

Libraries

pandas, NumPy, Requests, Beautiful Soup, SQLite, Plotly, Dash, dash-bootstrap-components, Jupyter.

AI tools

Claude (Anthropic) helped with the cleaning and dashboard code.